LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Practice Pack: Mixtures, Solutions and Separation

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. Level 1 checks the ideas, Level 2 is board-style questions, and Level 3 is Foundation-level challenge.
2. Mass by mass percentage = (mass of solute ÷ mass of solution) × 100, where solution = solute + solvent.

LEVEL 1: CONCEPTS

Q1. What is the difference between a homogeneous and a heterogeneous mixture? Give one example of each.

Q2. What is the Tyndall effect? In which kind of mixture is it seen?

Q3. Which method would you use to separate ammonium chloride from common salt?

Q4. Which method separates the different dyes in black ink?

Q5. Is boiling of water a physical or a chemical change? Is burning of paper?

LEVEL 2: BOARD-STYLE QUESTIONS

Q6. A solution contains 40 g of common salt in 320 g of water. Find its concentration as a mass by mass percentage.

Q7. How would you prepare 500 g of a 10% (by mass) sugar solution?

Q8. Give three differences between a true solution, a colloid and a suspension.

Q9. Which separation method would you use for each? (a) Cream from milk (b) Oil from water (c) Pure copper sulphate from an impure sample (d) Acetone from water

Q10. Give three differences between a mixture and a compound, using iron + sulphur as the example.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. How are the gases of air separated in industry? Why does this method work?

Q12. A student dissolved 5 g of urea in 95 g of water. A friend said the concentration is 5.26%. Who is right? Explain.

Q13. Why is crystallisation better than evaporation for getting pure crystals of a substance?

~ Match each separation method with the property it uses. That's the real trick! ~